

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405105
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Taira; Kenji

Method for operating an optical tomographic imaging apparatus

Abstract

A method for operating an optical tomographic imaging apparatus according to the present invention includes: an initial setting step of setting initial positions of a reference mirror and a distal end of an optical part; an imaging step of imaging a biological tubular element after the initial setting step; a reference mirror adjustment step of, after the imaging step, moving the reference mirror to enlarge the image portion of the reflected light from the biological tubular element and the image portion of the reflected light from the tube while reducing an image portion of an artifact caused by reflected light from the optical part, and adjusting the image portion of the artifact to an inside of the image portion of the reflected light from the tube; a magnification adjustment step of, after the reference mirror adjustment step, resetting the image portion of the reflected light from the biological tubular element and the image portion of the reflected light from the tube to a state before the enlargement; and a display step of, after the magnification adjustment step, causing an image display unit to display the image portion of the reflected light from the biological tubular element and the image portion of the reflected light from the tube reset to the state before enlargement.

Inventors:	Taira; Kenji (Musashino, JP)
Applicant:	GENESIS MEDTECH JAPAN CO., LTD. (Tokyo, JP)
Family ID:	80323503
Assignee:	GENESIS MEDTECH JAPAN CO., LTD. (Tokyo, JP)
Appl. No.:	18/111130
Filed:	February 17, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230213328 A1	Jul. 06, 2023

Foreign Application Priority Data

Related U.S. Application Data

continuation parent-doc WO PCT/JP2021/029914 20210816 PENDING child-doc US 18111130

Publication Classification

Int. Cl.: **G01B9/02091** (20220101); **A61B5/00** (20060101); **G01B9/02** (20220101);
G01B9/02015 (20220101)

U.S. Cl.:

CPC **G01B9/02091** (20130101); **A61B5/0066** (20130101); **G01B9/0203** (20130101);
G01B9/02084 (20130101);

Field of Classification Search

CPC: G01B (9/02091); G01B (9/0203); G01B (9/02084); G01B (9/02004); A61B (5/0066);
A61B (5/6852); A61B (1/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2012/0162660	12/2011	Kemp	N/A	N/A
2012/0250029	12/2011	Yoshida	356/497	G01B 9/02044
2015/0099984	12/2014	Kankaria	600/478	A61B 5/0066
2017/0224286	12/2016	Sakamoto	N/A	N/A
2018/0372477	12/2017	Elmaanaqui	N/A	N/A
2019/0223699	12/2018	Wu	N/A	A61B 1/00188
2019/0374109	12/2018	Wu	N/A	G06T 11/008
2021/0174125	12/2020	Zhang	N/A	G06T 7/12

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2018-164495	12/2015	JP	N/A
WO 2010/137373	12/2009	WO	N/A
WO 2018/047773	12/2015	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report for European Application No. 21858272.4, dated Jun. 11, 2024.
cited by applicant

English translation of International Preliminary Report on Patentability and Written Opinion of the
International Searching Authority for International Application No. PCT/JP2021/029914, dated
Mar. 2, 2023. cited by applicant

English translation of International Preliminary Report on Patentability for International

Application No. PCT/JP2020/031214, dated Feb. 23, 2023. cited by applicant
English translation of International Search Report for International Application No. PCT/JP2020/031214. dated Oct. 13, 2020. cited by applicant
English translation of International Search Report for International Application No. PCT/JP2021/029914, dated Oct. 19, 2021. cited by applicant

Primary Examiner: LaPage; Michael P

Attorney, Agent or Firm: Birch, Stewart, Kolasch & Birch, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of International Application No. PCT/JP2021/029914, with an International filing date of Aug. 16, 2021, which claims priority of International Application No. PCT/JP2020/031214 filed on Aug. 19, 2020, the entire content of which is hereby incorporated by reference.

TECHNICAL FIELD

(1) The present invention relates to a method for operating an optical tomographic imaging apparatus.

BACKGROUND ART

(2) Patent Document 1 (WO 2016/047773 A1) discloses a Swept Source Optical Coherence Tomography (SS-OCT) as a tomographic imaging system for imaging tomographic images of biological tissues such as various biological tubular elements including, for example, a gastrointestinal tract, a pancreaticobiliary duct, a fallopian tube, a urethra, a trachea, a blood vessel, and a lymph duct. This SS-OCT includes a wavelength swept light source that sequentially switches a wavelength and emits light, a reference mirror, and an optical probe that irradiates a biological tubular element with the light emitted from the wavelength swept light source. At a time of imaging, the light emitted from the wavelength swept light source is radiated toward the reference mirror and the optical probe in a state where the optical probe is inserted into a blood vessel to obtain interference light of reference mirror reflected light reflected by the reference mirror, and blood vessel reflected light reflected by the blood vessel through the optical probe. By detecting this interference light and performing Fourier transform, it is possible to obtain a tomographic image in a radial direction of the blood vessel.

(3) However, a plurality of optical parts are incorporated in the optical probe and optically coupled, light is reflected or scattered at boundary surfaces (e.g., light incident surfaces and light emission surfaces of the optical parts) of these optical parts, and this reflected light or scattered light may appear as an artifact in a tomographic image. There is a risk that the artifact not only lowers clarity of the image, but also make it difficult to perform correct diagnosis when the artifact overlaps a tissue image.

SUMMARY OF THE INVENTION

(4) Therefore, the present invention provides a method for preventing an artifact from overlapping an image of a biological tubular element in a tomographic image.

Means for Solving the Problems

(5) In order to achieve this object, a method for operating an optical tomographic imaging apparatus is provided, the optical tomographic imaging apparatus comprising:

(6) a light source;

(7) a splitter that splits light emitted from the light source into first light and second light;

(8) a translucent tube that is inserted into a biological tubular element;

- (9) an imaging unit that includes an optical fiber having a proximal end optically coupled to the splitter, and an optical part provided at a distal end of the optical fiber, and that emits the first light guided by the optical fiber from the optical part toward an inner wall of the biological tubular element through the tube, and obtain reflected light of the first light returning from the biological tubular element via the tube, from the optical part through the optical fiber;
- (10) an optical distance adjustment unit that includes a movable reference mirror, and is capable of obtaining reference light by reflecting the second light on the reference mirror, and adjusting an optical distance of the second light by moving the reference mirror;
- (11) an interference unit that causes the reflected light and the reference light to interfere with each other, and obtains interference light;
- (12) a detection unit that detects the interference light of the reflected light and the reference light;
- (13) a conversion unit that converts the interference light detected by the detection unit into an electrical signal;
- (14) a Fourier transform unit that performs Fourier transform on the electrical signal obtained by the conversion unit, and obtains a light intensity distribution with respect to an optical distance difference between an optical distance of the reflected light and an optical distance of the reference light;
- (15) an image processing unit that deletes an image portion of an artifact caused by the optical part from an image imaged by the image imaging unit, and obtains an image without the artifact;
- (16) an image display unit; and
- (17) a control unit that controls the imaging unit, the optical distance adjustment unit, the image processing unit, and the image display unit,
- (18) the method comprising executing, by the control unit:
- (19) (a) an initial setting step of controlling the optical distance adjustment unit to set an optical distance of the reference light substantially equally to an optical distance of the reflected light, the optical distance of the reference light being an optical distance that light emitted from the light source travels until reaching the detection unit after being reflected by the reference mirror, and the optical distance of the reflected light being an optical distance obtained by adding a first optical distance from the light source to a distal end of the optical part and a second optical distance from the distal end of the optical part to the detection unit;
- (20) (b) an imaging step of, after the initial setting step, operating the imaging unit to image the biological tubular element;
- (21) (c) a reference mirror adjustment step of, after the imaging step, controlling the optical distance adjustment unit to move the reference mirror to make the optical distance of the reference light shorter than the optical distance of the reflected light, thereby enlarge the image portion of reflected light from the biological tubular element and the image portion of reflected light from the tube while reducing an image portion of an artifact caused by the reflected light from the optical part, and fitting the image portion of the artifact to an inside of the image portion of the reflected light from the tube;
- (22) (d) a magnification adjustment step of, after the reference mirror adjustment step, controlling the image processing unit to reset the image portion of the reflected light from the biological tubular element and the image portion of the reflected light from the tube to a state before the enlargement; and
- (23) (e) a display step of, after the magnification adjustment step, causing the image display unit to display the image portion of the reflected light from the biological tubular element and the image portion of the reflected light from the tube reset to the state before the enlargement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus are not limitative of the present invention, and wherein:
- (2) FIG. 1 is a view illustrating a schematic configuration of an optical tomographic imaging device according to an embodiment of the present invention.
- (3) FIG. 2 is a cross-sectional view of an optical probe included in the optical tomographic imaging device illustrated in FIG. 1.
- (4) FIG. 3A illustrates an intensity distribution of an OCT signal obtained by the optical tomographic imaging device illustrated in FIG. 1.
- (5) FIG. 3B illustrates an intensity distribution of an OCT signal obtained by the optical tomographic imaging device illustrated in FIG. 1.
- (6) FIG. 3C illustrates an intensity distribution of an OCT signal obtained by the optical tomographic imaging device illustrated in FIG. 1.
- (7) FIG. 4A is a view illustrating a tomographic image of a biological tubular element including an artifact caused by an optical part.
- (8) FIG. 4B is a view illustrating a state where the artifact is made small in the tomographic image of the biological tubular element.
- (9) FIG. 4C is a view illustrating a tomographic image of the biological tubular element obtained by removing the artifact from FIG. 4A.

EMBODIMENTS OF THE INVENTION

- (10) Hereinafter, embodiments of an optical tomographic imaging method according to the present invention and an optical tomographic imaging device that uses this method will be described with reference to the accompanying drawings.
- (11) [Optical Tomographic Imaging Device]
- (12) FIG. 1 illustrates an outline of a Swept Source Optical Tomograph (SS-OCT) device according to the embodiment.
- (13) An SS-OCT **100** includes an optical unit **200** and a signal processing unit **400**.
- (14) [Optical Unit]
- (15) The optical unit **200** includes a plurality of optical elements. The plurality of optical elements include a wavelength swept light source **210**, a first optical coupler **212**, a first optical circulator **214**, a second optical circulator **216**, a second optical coupler **218**, a biological tubular element imaging unit **220**, an optical distance adjustment unit **222**, and a detection unit **224**. The relevant elements of these optical elements are optically coupled by light transmission elements such as optical fibers as described below.
- (16) [Wavelength Swept Light Source]
- (17) The wavelength swept light source **210** outputs light necessary for imaging a cross-section of a biological tubular element. The wavelength swept light source **210** is configured to be able to cyclically change a wavelength of light to be output, and sweep a wavelength from 1260 nm to 1360 nm at, for example, 100 kHz in frequency.
- (18) [First Optical Coupler]
- (19) The first optical coupler (splitter) **212** is a portion that is optically coupled by heating and fusing part of two optical fibers disposed in parallel. One of the two optical fibers optically couples the first optical coupler **212** and the first optical circulator **214**, and the other optical fiber optically couples the wavelength swept light source **210** and the second optical circulator **216** with the first optical coupler **212** interposed therebetween. Therefore, light output from the wavelength swept light source **210** is split into two light beams by the first optical coupler **212**, the one split light beam is sent to the first optical circulator **214**, and the other split light beam is sent to the second optical circulator **216**.

(20) [First Optical Circulator]

(21) The first optical circulator **214** is a 3-port optical circulator, includes a first port coupled to the first optical coupler **212**, a second port coupled to the second optical coupler **218**, and a third port coupled to the biological tubular element imaging unit **220**, and is configured to send to the biological tubular element imaging unit **220** the light sent from the wavelength swept light source **210** to the first optical circulator **214** through the first optical coupler **212**, and send to the second optical coupler **218** light returned from the biological tubular element imaging unit **220**.

(22) [Second Optical Circulator]

(23) The second optical circulator **216** is a 3-port optical circulator, includes a first port coupled to the first optical coupler **212**, a second port coupled to the second optical coupler **218**, and a third port coupled to the optical distance adjustment unit **222**, and is configured to send to the optical distance adjustment unit **222** the light sent from the wavelength swept light source **210** to the second optical circulator **216** through the first optical coupler **212**, and send to the second optical coupler **218** light returned from the optical distance adjustment unit **222**.

(24) [Second Optical Coupler]

(25) At a portion at which an optical fiber having one end (proximal end) connected to the first optical circulator **214** and a middle of an optical fiber having one end (proximal end) connected to the second optical circulator **216** are heated and fused, and optically coupled, the second optical coupler **218** (interference unit) overlaps light (reflected light) sent from the biological tubular element imaging unit **220** to the second optical coupler **218** through the first optical circulator **214**, and light (reference light) sent from the optical distance adjustment unit **222** to the second optical coupler **218** through the second optical circulator **216** to obtain interference light.

(26) [Biological Tubular Element Imaging Unit]

(27) The biological tubular element imaging unit **220** includes a base **226**, and a linear motion unit (pull-back unit) **229** that linearly moves in a predetermined direction (a left-right direction in FIG. 1) with respect to the base **226**. The linear motion unit **228** is coupled to a linear motion motor **230** provided to the base **226**, and is configured to move forward or backward in the predetermined direction based on driving of the linear motion motor **230**. To the base **226**, a proximal end part of a hollow cylindrical flexible tube (referred to as a sheath below) **232** made of a translucent resin is detachably fixed by a retainer **233**. As illustrated, the sheath **232** includes an opened proximal end, and a closed distal end.

(28) The linear motion unit **228** is provided with a proximal end side collimation lens **234** and a distal end side collimation lens **236**. The proximal end side collimation lens **234** and the distal end side collimation lens **236** are aligned at a fixed interval on one optical axis, and light having passed through the proximal end side collimation lens **234** is transmitted on the optical axis of the distal end side collimation lens **236**.

(29) The proximal end side collimation lens **234** is fixed to the linear motion unit **228**, and is optically coupled to the first optical circulator **214** with the optical fiber interposed therebetween.

(30) The distal end side collimation lens **236** is supported by a rotation unit **233** provided to the linear motion unit **228**. In a state where a fixed distance between the proximal end side collimation lens **234** and the distal end side collimation lens **236** is kept, the rotation unit **238** is supported by the linear motion unit **229** rotatably about the optical axes of the proximal end side collimation lens **234** and the distal end side collimation lens **236**.

(31) The rotation unit **238** is drivingly coupled to a rotation motor **240** fixed to the linear motion unit **228** with a rotation transmission mechanism (not illustrated) interposed therebetween and including, for example, a gear and a toothed belt.

(32) An optical probe **300** is detachably coupled to the rotation unit **234**. The optical probe **300** includes an optical fiber **310** and an optical part **312**. The optical part **312** is coupled to a distal end of the optical fiber **310**. A proximal end of the optical fiber **310** is detachably coupled to the rotation unit **238** with a connector **241** interposed therebetween, and the optical fiber **310** is

configured to rotate together with rotation of the rotation unit **238** such that light condensed by the distal end side collimation lens **236** is incident on a core of the optical fiber **310** from the proximal end of the optical fiber **310**.

(33) The optical part **312** coupled to the distal end of the optical fiber **310** includes a plurality of optical elements. In the embodiment, the plurality of optical elements include a glass rod (first lens) **314**, a grin lens (second lens) **316**, and, in addition, a prism **318** that is a deflection part (see FIG. 2). The glass rod **314**, the grin lens **316**, and the prism **318** are disposed in this order from the proximal end side toward the distal end side, and the adjacent optical elements are mechanically and optically coupled to each other by fusion or an adhesive of a translucent resin.

(34) The glass rod **314** that has a cylindrical shape and the grin lens **316** that is a cylindrical lens are installed coaxially with respect to an extension line (optical axis) of the optical axis of the optical fiber **310**. A substantially right angle prism having a triangular prism shape is used for the prism **314** that is a deflection part, and one side (surface) of two sides (sides other than an oblique side) sandwiching the substantially right angle is coupled to a distal end surface of the grin lens **316**. Therefore, light emitted from the distal end of the optical fiber **310** enters the prism **318** from one side (surface) on the proximal end side through the glass rod **314** and the grin lens **316**. The light having entered the prism **318** is reflected by an inclined surface (oblique side), then travels in a direction orthogonal to the optical axis, and is emitted in a radial direction from the other side (surface) on the distal end side. Conversely, light having entered the prism **318** from the other side (surface) on the distal end side is reflected by the inclined surface (oblique side), then sequentially passes through the grin lens **316** and the glass rod **314** through one side (surface) on the proximal end side, and is incident on the optical fiber **310**.

(35) In order to suppress as much as possible the light reflected by end surfaces of the optical fiber **310** and the optical part **312** and returned to the optical fiber **310** among the light sent through the optical fiber **310**, a distal end surface of the optical fiber **310**, proximal end side end surfaces and distal end side end surfaces of the glass rod **314** and the grin lens **316**, and a proximal end surface of the prism **318** are preferably inclined slightly with respect to a plane perpendicular to the optical axis (e.g., approximately 8° to approximately 10°).

(36) In order to suppress the light emitted from the prism **318** from spreading in an optical axis direction, the inclined surface of the prism **31e** may be formed as, for example, a curved surface whose center is an axis orthogonal to the optical axis and that is convex outward. In order to suppress the light emitted from the prism **318** from spreading in a circumferential direction around the optical axis, the other side (surface) on the distal end side of the prism **318** may be formed as, for example, a curved surface whose center is an axis parallel to the optical axis of light propagating through the optical fiber **310** and that is convex outward.

(37) The above-described sheath **232** and optical probe **300** are provided in a state where the optical probe **300** is inserted into the sheath **232**. At a time of use, the optical fiber **310** of the optical probe **300** is coupled to the rotation unit **238** with the connector **241** interposed therebetween, and the proximal end part of the sheath **232** is fixed to the base **226** by the retainer **233**.

(38) Therefore, the linear motion unit **228** and the optical probe **300** move forward and backward with respect to the base **226** based on driving of the linear motion motor **230**. On the other hand, the sheath **232** is held by the base **226**. Therefore, the optical probe **300** moves forward and backward in the sheath **232** based on driving of the linear motion motor **230**. Furthermore, the rotation unit **238** rotates about the optical axis based on driving of the rotation motor **240**. At this time, the optical probe **300** is coupled to the rotation unit **238** with the connector **241** interposed therebetween, and therefore rotates together with the rotation unit **238**. Accordingly, the light sent from the first optical circulator **214** to the imaging unit **220** is incident on the optical fiber **310** through the proximal end side collimation lens **234** and the distal end side collimation lens **236**, and is then emitted in the radial direction from the optical fiber **310** through the optical part **312**, and

this emission light is scanned at a constant speed in the circumferential direction whose center is the optical axis.

(39) Sizes of the sheath **232** and the optical probe **300** accommodated in the sheath **232** are appropriately determined according to a size of a biological tubular element to be imaged. An inner diameter of the sheath **232** and an outer diameter of the optical probe **300** accommodated therein are determined to be, for example, approximately 50 μm larger than a maximum outer diameter of the glass rod **314**, the grin lens **316**, the prism **318**, and a metal pipe (not illustrated) that holds these optical elements such that the optical probe **300** stably rotates in the sheath **232**. In a case where, for example, the maximum outer diameter of the glass rod **314**, the grin lens **316**, the prism **318**, and the metal pipe holding these optical elements is approximately 200 to 500 μm , the inner diameter of the sheath **232** is approximately 250 to 550 μm .

(40) [Optical Distance Adjustment Unit]

(41) The optical distance adjustment unit **222** includes a collimation lens **242** and a reference mirror **244**. The collimation lens **242** is immovably fixed. The reference mirror **244** includes on the optical axis of the collimation lens **242** a reflection surface (mirror surface) that is vertical to the optical axis. The reference mirror **244** is supported by the linear motion unit **246**. The linear motion unit **246** is movable along the optical axis of the collimation lens **242**. The linear motion unit **246** is also coupled to the linear motion motor **248**, and is configured to move forward and backward toward the collimation lens **242** based on driving of the linear motion motor **248**, and thereby adjust an optical distance (optical distance) of reference light.

(42) [Detection Unit]

(43) The detection unit **224** is an optical part that receives light sent from the second optical coupler **218** and photoelectrically converts the light, and, more particularly, is a dual balanced detector that includes two light input units. The dual balanced detector includes two photodiodes that detect light (interference light) sent from the second optical coupler **218**, and these two photodiodes are respectively coupled to distal ends of two optical fibers that make up the second optical coupler **218**. The detection unit **224** is also configured to convert the light beams input from the two optical fibers into electric signals, then make direct current components included in the electric signals cancel each other, and thereby extract only the electric signals that are based on the interference light.

(44) [Signal Processing Unit]

(45) The signal processing unit **400** includes a control unit **410**, an image adjustment unit **412**, an analog-digital (A/D) conversion unit **414**, a Fourier transform unit **416**, an image processing unit (artifact deletion unit) **418**, and an image display unit (monitor) **420**. In the drawing, the image adjustment unit **412**, the A/D conversion unit **414**, the Fourier transform unit **416**, and the image processing unit **418** indicate functional blocks, and do not need to have physical components, and may be part of a program that is installed in a storage unit of the control unit **410** described later.

(46) [Control Unit]

(47) Although not illustrated, the control unit **410** includes a controlling unit, an arithmetic operation unit, and a storage unit. The storage unit can temporarily store a program that is necessary to execute processing to be described later, and various items of data (e.g., image data) generated in a process of the processing to be described later. The arithmetic operation unit executes an arithmetic operation that needs to be executed in the process of the processing to be described later according to the program stored in the storage unit.

(48) Although not illustrated, the control unit **410** is communicably connected with various devices (the wavelength swept light source **210**, the optical probe rotation motor **240**, the optical probe linear motion motor **230**, and the reference mirror linear motion motor **249**) included in the optical unit **200**, and is configured to individually drive and control these devices according to the program of the storage unit. The control unit **410** is also communicably connected with the image adjustment unit **412**, the analog-digital (A/D) conversion unit **414**, the Fourier transform unit **416**,

the image processing unit (artifact deletion unit) **418**, and the image display unit (monitor) **420**, and is configured to exchange signals with these components.

(49) Although not illustrated, the control unit **410** is connected to the input unit, and is configured to execute processing to be described below based on a signal input from the input unit. The input unit may be any of a keyboard, a pointing device, a touch screen, a mouse, a joystick, a trackball, a scanner, an OCR, an OMR, an audio input device, a tablet, and the like.

(50) [Image Adjustment Unit]

(51) The image adjustment unit **412** moves the reference mirror **244** by driving the linear motion motor **248** of the optical distance adjustment unit **222**, and adjusts the optical distance of the reference light. In the embodiment, in an initial state, the image adjustment unit **412** sets the optical distance that the light emitted from the light source **210** travels until the light is reflected by the reference mirror **244** and reaches the detection unit **224** equally to an optical distance obtained by calculating a sum of the optical distance from the light source **210** to the prism inclined surface **320** and the optical distance from the prism inclined surface **320** to the detection unit **224**.

(52) A position of the reference mirror **244** in the initial state will be referred to as a “reference position” below. Therefore, when, for example, the reference mirror **244** is moved toward the collimation lens **242**, a point (referred to as a “post-movement corresponding point **330**” below) on an optical path of the reflected light corresponding to the position of the reference mirror **244** after the movement moves to a front side of the inclined surface **320** of the prism **318** (see FIG. 2). As a result, as will be described later, when the reference mirror **244** is moved toward the collimation lens **242**, while distances from the post-movement corresponding point **330** on the optical path of the reflected light to the sheath **232** and a biological tubular element **500** become long, distances from the post-movement corresponding point **330** to bonding surfaces of the optical elements (bonding surfaces of the glass rod **314**, the grin lens **316**, and the prism **318**) become short.

(53) [A/D Conversion Unit]

(54) The A/D conversion unit **414** converts an analog signal (an electric signal based on interference light) output from the detection unit **224** into a digital signal.

(55) [Fourier Transform Unit]

(56) The Fourier transform unit **416** performs Fourier transform (e.g., fast Fourier transform or discrete Fourier transform) on the digital signal output from the A/D conversion unit **414**, and obtain an intensity (power spectrum) of the interference light with respect to the difference (optical distance difference) between the optical distance of the reflected light and the optical distance of the reference light.

(57) FIG. 3A illustrates a relationship between an optical distance difference and an intensity obtained when the optical probe **300** covered with the sheath **232** is inserted into the biological tubular element **500** (e.g., blood vessel) as illustrated in FIG. 2. In this FIG. 3A, a horizontal axis indicates the optical distance difference on a linear scale, a vertical axis indicates the intensity on a logarithmic scale, an intensity distribution denoted by reference numeral **610** corresponds to interference light of reflected light from the biological tubular element **500** and reference light, and an intensity distribution denoted by reference numeral **612** corresponds to interference light of reflected light from the sheath **232** and reference light. In FIG. 3A, an intensity distribution denoted by reference numeral **614** corresponds to interference light (above-described artifact **700A**) of light reflected or scattered on or near the bonding surfaces of the optical elements (bonding surfaces of the glass rod **314**, the grin lens **316**, and the prism **318**) that make up the optical probe **300** and reference light. Although the intensity distribution **610** of the biological tubular element **500** and the intensity distribution **614** of the artifact **700A** are separately displayed in FIG. 3A for ease of understanding, an actually obtained power spectrum appears in a form obtained by synthesizing the two intensity distributions **610** and **614**. Furthermore, in FIG. 3A, the intensity distribution **610** of the biological tubular element **500** and the intensity distribution **614** of the artifact **700A** appear (overlap) in the same optical distance difference area because the optical distance from the prism

inclined surface **320** (reference surface) to the biological tubular element **500** and the optical distance from the prism inclined surface **320** to the bonding surfaces of the optical elements (the bonding surfaces of the glass rod **314**, the grin lens **316**, and the prism **318**) are substantially equal. (58) [Image Processing Unit]

(59) The image processing unit **418** receives a power spectrum including the spectral components **610** and **612** of the biological tubular element **500** and the sheath **232** and the spectral component **614** of an artifact **700** from the Fourier transform unit **416**, and outputs image information per rotation angle of the optical probe **300** according to the intensity distributions **610**, **612**, and **614** of the biological tubular element **500**, the sheath **232**, and the artifact **700** in the power spectrum.

(60) The image processing unit **418** also deletes an image portion (more specifically, the artifact) of an instructed area in the captured image based on an instruction from the control unit **410** as described in detail later.

(61) [Image Display Unit]

(62) The image display unit **420** is a general monitor.

(63) For reference, FIG. **4A** illustrates a captured image obtained when the optical probe **300** covered with the sheath **232** is inserted into the biological tubular element **500** (e.g., blood vessel) as illustrated in FIG. **2**. In FIG. **4A**, an image portion denoted by reference numeral **500A** corresponds to the biological tubular element **500**, an image portion denoted by reference numeral **232A** corresponds to the sheath **232**, and an image portion (referred to as an “artifact” below) denoted by reference numeral **700A** corresponds to light reflected or scattered on or near the bonding surfaces of the optical elements.

(64) [Optical Tomographic Imaging]

(65) Optical tomographic imaging that uses the above-described SS-OCT **100** will be described.

(66) The optical probe **300** is inserted together with the sheath **232** into an imaging site of a biological tubular element (a gastrointestinal tract, a pancreaticobiliary duct, a fallopian tube, a urethra, a trachea, a blood vessel, or a lymph duct) of a human or an animal. When, for example, a cross section of a blood vessel of a human is imaged, a guide wire is inserted from a basilic vein of the upper arm, and then the optical probe **300** and the sheath **232** are inserted to a target site along a guide wire.

(67) When the optical probe **300** is moved, an operator inputs a necessary signal to the control unit **410** through an unillustrated input unit. The control unit **410** that has received this signal drives the linear motion motor **230** of the biological tubular element imaging unit **220** to move the linear motion unit **226** forward or backward.

(68) Next, the control unit **410** drives the rotation motor **240** of the biological tubular element imaging unit **220** based on a driving start signal from the unillustrated input unit to rotate the optical probe **300** at a speed of, for example, 180 rps.

(69) Next, the control unit **410** drives the wavelength swept light source **210** to emit light of a predetermined wavelength. The light emitted from the wavelength swept light source **210** is split into first light traveling to the biological tubular element imaging unit **220**, and second light traveling to the optical distance adjustment unit **222** in the first optical coupler **212**.

(70) The split first light is sent to the biological tubular element imaging unit **220** via the first optical circulator **214**, and is incident on the optical fiber **310** of the optical probe **300** through the proximal end side collimation lens **234** and the distal end side collimation lens **236**. The light incident on the optical fiber **310** passes through the core of the optical fiber **310**, then reaches the inclined surface **320** (reference surface) of the prism **318** via the glass rod **314** and the grin lens **316**, and is reflected by the inclined surface **320**. The light reflected by the inclined surface **320** of the prism **318** is emitted in the radial direction from a prism outer circumferential surface **322**. The light emitted from the prism **318** is transmitted through the sheath **232** and radiated on an entire circumference of an inner wall of the biological tubular element **500** while moving in the circumferential direction based on rotation of the optical probe **300**. The light reflected by the

biological tubular element **500** and biological tissues around the biological tubular element **500** enters the prism **318** through the outer circumferential surface **322** of the prism **318**, is reflected by the inclined surface **320**, and then is sent to the first optical circulator **214** through the grin lens **316**, the glass rod **314**, the optical fiber **310**, the distal end side collimation lens **236**, and the proximal end side collimation lens **234**. The first optical circulator **214** sends the reflected light from the biological tubular element imaging unit **220** toward the second optical coupler **218**.

(71) The reflected light sent to the first optical circulator **214** includes the reflected light from the biological tubular element **500** and the sheath **232**, and, in addition, unnecessary reflected light and scattered light from the bonding surfaces of the glass rod **314**, the grin lens **316**, and the prism **318** (see FIGS. **3A** and **4A**).

(72) The second light split by the first optical coupler **212** is sent from the second optical circulator **216** to the optical distance adjustment unit **222**, emitted from the collimation lens **242**, then reflected by the reference mirror **244** facing this collimation lens **242**, and sent again to the second optical circulator **216** through the collimation lens **242**. The second optical circulator **216** sends the reference light from the optical distance adjustment unit **222** toward the second optical coupler **218**.

(73) The reflected light sent to the second optical coupler **218** and the reference light are synthesized to obtain interference light. The interference light is split into two light beams, and each light beam is incident on the detection unit **224**.

(74) The detection unit **224** photoelectrically converts the interference light to generate an analog signal corresponding to the interference light. The generated analog signal is sent to the A/D conversion unit **414**, and converted into a digital signal.

(75) FIG. **4A** illustrates an image (an unprocessed image including an artifact) created based on the digital signal. As illustrated in FIG. **4A**, the unprocessed image includes the image portion **500A** of the biological tubular element **500**, the image portion **232A** of the sheath **232**, and, in addition, and, in addition, the image portion (artifact) **700A** of light reflected or scattered on or near the bonding surfaces of the optical elements of the optical probe **300**. In the embodiment, the distance from the prism inclined surface (reference surface) **320** to the biological tubular element **500** and the distance from the prism inclined surface (reference surface) **320** to the bonding surfaces of the optical elements (the bonding surfaces of the glass rod **314**, the grin lens **316**, and the prism **318**) are set substantially equally. Hence, the artifact **700A** overlaps the image portion **500A** of the biological tubular element **500** of most interest, which makes the image of the image portion **500A** unclear.

(76) As described above, the digital signal including information of the artifact **700A** is sent to the Fourier transform unit **416**. The Fourier transform unit **416** performs Fourier transform (fast Fourier transform or discrete Fourier transform) on the digital signal. As a result, as illustrated in FIG. **3A**, an intensity distribution (power spectrum) of the interference light corresponding to the optical distance difference between the optical distance of the reference light and the optical distance of the reflected light is obtained. As described above, the distance from the prism inclined surface (reference surface) **320** to the biological tubular element **500** and the distance from the prism inclined surface (reference surface) **320** to the bonding surfaces of the optical elements are substantially equal, and therefore the intensity distribution **614** of the artifact **700A** appears in substantially the same optical distance difference area as that of the intensity distribution **610** of the biological tubular element **500**. Therefore, as illustrated in FIG. **4A**, the artifact **700A** appears overlapping the image **500A** of the biological tubular element **500**.

(77) In order to remove the artifact **700A** and make the image **500A** of the biological tubular element **500** clear, the control unit **410** drives the linear motion motor **248** of the optical distance adjustment unit **222** via the image adjustment unit **412** to bring the reference mirror **244** close to the collimation lens **242**. Thus, the optical distance that the reference light reflected by the reference mirror **244** travels until reaching the detection unit **224** becomes short. As a result, while the difference between the optical distance of the reflected light from the biological tubular element

500 and the sheath **232** and the optical distance of the reference light becomes great, the difference between the optical distance of the reflected light or the scattered light from the bonding surface of the optical part **312** in the optical probe **300** and the optical distance of the reference light becomes a little. Hence, as illustrated in FIG. **4B**, while an image portion **500B** of the biological tubular element **500** and an image portion **232B** of the sheath **232** are enlarged, an image portion **700B** of the artifact is reduced, and the image portion **700B** of the artifact is fitted to an inside of the image portion **232B** of the sheath **232**. Therefore, as illustrated in FIG. **3B**, in the power spectrum obtained by performing Fourier transform on the interference light after the optical distance adjustment, the intensity distributions **610** and **612** of the biological tubular element **500** and the sheath **232** move to a right side in FIG. **3B**, and the intensity distribution **614** of the artifact caused by the light reflected or scattered on or near the bonding surface of the optical element moves to a left side in FIG. **3B**, such that the intensity distributions **610** and **612** and the intensity distribution **614** are separated so as not to overlap each other.

(78) The distance that the reference mirror **244** moves will be described. In the SS-OCT **100**, a size (an inner diameter and an outer diameter) of the sheath **232** and sizes (lengths in the optical axis direction) of the optical elements (the glass rod **314**, the grin lens **316**, and the prism **318**) of the optical probe **300** are determined according to models of the sheath **232** and the probe **300**, and information of these sizes is stored in the storage unit of the control unit **410**. Therefore, according to models of the sheath **232** and the probe **300**, the control unit **410** determines a movement distance of the reference mirror **244** such that the optical distance to the sheath **232** from the post-movement corresponding point **330** (FIG. **2**) on the optical path of the reflected light corresponding to the position of the reference mirror **244** after movement is longer than the optical distance from the post-movement corresponding point **330** to the bonding surfaces of the optical elements such as the bonding surfaces of the grin lens **316** and the prism **318**. This movement distance may be stored according to the models of the sheath **232** and the probe **300**, and the control unit **410** may move the reference mirror **244** based on stored values.

(79) Next, the control unit **410** activates the image processing unit **419**, and deletes the image portion **700B** of the reduced artifact from the image illustrated in FIG. **4B**. More specifically, the control unit **410** deletes the image data inside the annular image portion **232B** of the sheath **232** (the image portion **700B** corresponding to the artifact), that is, image data within an optical distance difference range including the intensity distribution **614** in FIG. **3B**. The optical distance difference range corresponding to the image data that needs to be deleted corresponds to the movement distance of the above-described reference mirror **244**, is determined in advance according to the models of the sheath **232** and the probe **300** together with the movement distance, and is stored in the storage unit.

(80) Subsequently, the control unit **420** resets the enlarged image (see FIG. **4B**) from which the image data of the artifact has been deleted to a magnification of the image before enlargement illustrated in FIG. **4A**. FIG. **4C** illustrates an image whose magnification has been reset, and that includes no artifact.

(81) Finally, the control unit **410** displays the image illustrated in FIG. **4C** on the display unit **420**. As described above, the image portion of the artifact is removed from the image displayed on the display unit **420**. Consequently, it is possible to obtain a clear biological tubular element tomographic image without the artifact. As a result, reliability of the image to be obtained increases, and reliability of diagnosis that uses this image increases.

(82) Note that, although FIG. **4A** illustrates the image including the artifact, and FIG. **4B** illustrates the image enlarged from the image of FIG. **4A** in the above description to facilitate understanding of the invention, an actual device does not display these images on the display unit **420**. Naturally, the images in FIGS. **4A** and **4B** may be displayed on the display unit **420** to indicate presence of the artifact.

Other Embodiment

(83) Although the control unit **410** activates the image processing unit **418** to delete the reduced image portion **700B** of the artifact from the enlarged image illustrated in FIG. **4B** in the above-described embodiment, this reduced image portion **700B** of the artifact may not be deleted. In this case, even when the control unit **410** resets the enlarged image illustrated in FIG. **4B** to the magnification of the image before the enlargement illustrated in FIG. **4A**, the image portion **700B** of the artifact remains fitted inside the image portion **232B** of the sheath **232**, and therefore the image portion **500B** of the biological tubular element **500** and the image portion **700B** of the artifact do not overlap. Consequently, even if the reduced image portion **700B** of the artifact is not deleted, reliability of an image to be obtained increases, and reliability of diagnosis that uses this image increases.

(84) Furthermore, to continuously obtain images, the control unit **410** may perform a pull-back operation of moving the linear motion unit **229** backward while rotating the optical probe **300**. In this case, after displaying the image illustrated in FIG. **4C** on the display unit **420**, the control unit **410** operates the biological tubular element imaging unit **220** to image the biological tubular element **500** while moving the optical probe **300** backward along the sheath **232** by the pull-back operation. At this time, the position of the reference mirror **244** has already been adjusted, and therefore while the image portion of the biological tubular element **500** and the image portion of the sheath **232** are enlarged as described above, the image portion of the artifact is reduced and fitted inside the image portion of the sheath **232**. After imaging by this pull-back operation, the control unit **410** resets the enlarged image after the image data of the artifact is deleted or the enlarged image in which the image portion of the artifact remains fitted inside the image portion of the sheath **232** to the magnification of the image before enlargement by the same method as that described above, and then displays the image reset to the magnification of the image before enlargement on the display unit **420**. As described above, the control unit **410** can continuously obtain images by repeating imaging by the pull-back operation, the process of resetting the enlarged image to the magnification of the image before enlargement, and the process of displaying on the display unit **420** the image reset to the magnification of the image before enlargement.

(85) Although the optical probe **300** includes the glass rod **314**, the grin lens **316**, and the prism **318** in the above-described embodiments, the optical parts making up the optical probe **300** are not limited to these. An optical element may be formed in place of a prism by, for example, obliquely machining a distal end of a core of an optical fiber, and light may be radiated in the radial direction from an inclined surface of this distal end. Furthermore, the optical probe may include an optical fiber and a glass rod or a plastic rod including an inclined distal end surface. Alternatively, the optical probe may include an optical fiber and a ball-type lens including an inclined distal end surface.

(86) Although the optical distance of the reference light is set using the inclined surface **320** of the prism **318** as the reference surface in the above-described embodiments, the outer circumferential surface **322** of the prism **318** may be used as the reference surface.

(87) Furthermore, the image processing unit **418** may include a change amount detection unit (not illustrated) that detects an amount of change in an optical distance difference between the optical distance of reference light and an optical distance of reflected light when obtaining an image. This change amount detection unit automatically detects the amount of change in the optical distance difference between the optical distance of the reference light and the optical distance of the reflected light caused by, for example, expansion and contraction of the optical fiber **310** during imaging by the SS-OCT **100**, and sends an instruction that is based on the amount of change to the image adjustment unit **412** through the control unit **410**. The image adjustment unit **412** moves the reference mirror **244** based on above the instruction by driving the linear motion motor **240** of the optical distance adjustment unit **222**, and automatically adjusts the optical distance of the reference light. Consequently, the optical distance difference between the optical distance of the reference light and the optical distance of the reflected light is automatically maintained. Consequently, it is

possible to reliably delete an artifact from a tomographic image without changing the amount of change in the optical distance difference between the optical distance of the reference light and the optical distance of the reflected light during imaging.

(88) As described above, a method for operating an optical tomographic imaging apparatus is provided, the optical tomographic imaging apparatus comprising:

(89) a light source;

(90) a splitter that splits light emitted from the light source into first light and second light;

(91) a translucent tube that is inserted into a biological tubular element;

(92) an imaging unit that includes an optical fiber having a proximal end optically coupled to the splitter, and an optical part provided at a distal end of the optical fiber, and that emits the first light guided by the optical fiber from the optical part toward an inner wall of the biological tubular element through the tube, and obtain reflected light of the first light returning from the biological tubular element via the tube, from the optical part through the optical fiber;

(93) an optical distance adjustment unit that includes a movable reference mirror, and is capable of obtaining reference light by reflecting the second light on the reference mirror, and adjusting an optical distance of the second light by moving the reference mirror;

(94) an interference unit that causes the reflected light and the reference light to interfere with each other, and obtains interference light;

(95) a detection unit that detects the interference light of the reflected light and the reference light;

(96) a conversion unit that converts the interference light detected by the detection unit into an electrical signal;

(97) a Fourier transform unit that performs Fourier transform on the electrical signal obtained by the conversion unit, and obtains a light intensity distribution with respect to an optical distance difference between an optical distance of the reflected light and an optical distance of the reference light;

(98) an image processing unit that deletes an image portion of an artifact caused by the optical part from an image imaged by the image imaging unit, and obtains an image without the artifact;

(99) an image display unit; and

(100) a control unit that controls the imaging unit, the optical distance adjustment unit, the image processing unit, and the image display unit,

(101) the method comprising executing, by the control unit:

(102) (a) an initial setting step of controlling the optical distance adjustment unit to set an optical distance of the reference light substantially equally to an optical distance of the reflected light, the optical distance of the reference light being an optical distance that light emitted from the light source travels until reaching the detection unit after being reflected by the reference mirror, and the optical distance of the reflected light being an optical distance obtained by adding a first optical distance from the light source to a distal end of the optical part and a second optical distance from the distal end of the optical part to the detection unit;

(103) (b) an imaging step of, after the initial setting step, operating the imaging unit to image the biological tubular element;

(104) (c) a reference mirror adjustment step of, after the imaging step, controlling the optical distance adjustment unit to move the reference mirror to make the optical distance of the reference light shorter than the optical distance of the reflected light, thereby enlarge the image portion of reflected light from the biological tubular element and the image portion of reflected light from the tube while reducing an image portion of an artifact caused by the reflected light from the optical part, and fitting the image portion of the artifact to an inside of the image portion of the reflected light from the tube;

(105) (d) a magnification adjustment step of, after the reference mirror adjustment step, controlling the image processing unit to reset the image portion of the reflected light from the biological tubular element and the image portion of the reflected light from the tube to a state before the

enlargement; and

(106) (e) a display step of, after the magnification adjustment step, causing the image display unit to display the image portion of the reflected light from the biological tubular element and the image portion of the reflected light from the tube reset to the state before the enlargement.

(107) According to the method according to embodiments of the present invention, an artifact caused by an optical part does not overlap a tomographic image of the biological tubular element. Consequently, it is possible to appropriately perform diagnosis that uses tomographic images.

(108) It is to be noted that the various embodiments described above can be appreciated individually within each embodiment, but the embodiments can be combined together. It is also to be noted that the various features in different embodiments can be appreciated individually by its own, but the features in different embodiments can be combined.

Claims

1. A method for operating an optical tomographic imaging apparatus, the optical tomographic imaging apparatus comprising: a light source; a splitter that splits light emitted from the light source into first light and second light; a translucent tube that is inserted into a biological tubular element; an imaging unit that includes an optical fiber having a proximal end optically coupled to the splitter, and an optical part provided at a distal end of the optical fiber, and that emits the first light guided by the optical fiber from the optical part toward an inner wall of the biological tubular element through the translucent tube, and obtain reflected light of the first light returning from the biological tubular element via the translucent tube, from the optical part through the optical fiber; an optical distance adjustment unit that includes a movable reference mirror, and is capable of obtaining reference light by reflecting the second light on the reference mirror, and adjusting an optical distance of the second light by moving the reference mirror; an optical coupler that causes the reflected light and the reference light to interfere with each other, and obtains interference light; a detector that detects the interference light of the reflected light and the reference light and photoelectrically converts the interference light to generate an analog signal corresponding to the interference light; a converter that receives the analog signal from the detector and converts the analog signal into a digital signal; a Fourier transform unit that performs Fourier transform on the digital signal obtained by the converter to obtain a light intensity distribution chart with respect to an optical distance difference between an optical distance of the reflected light and an optical distance of the reference light; an image display; an image processor that deletes data corresponding to an intensity corresponding to the artifact from the digital data Fourier-transformed by Fourier transform unit, so that an image without the artifact is displayed on the image display; and a controller, wherein the imaging unit, the optical distance adjustment unit, the image processor, and the image display are configured to be controlled by the controller, the method comprising executing, by the controller: (a) an initial setting step of controlling the optical distance adjustment unit to set an optical distance of the reference light substantially equally to an optical distance of the reflected light, the optical distance of the reference light being an optical distance that light emitted from the light source travels until reaching the detector after being reflected by the reference mirror, and the optical distance of the reflected light being an optical distance obtained by adding a first optical distance from the light source to a distal end of the optical part and a second optical distance from the distal end of the optical part to the detector; (b) an imaging step of, after the initial setting step, operating the imaging unit to image the biological tubular element through the translucent tube by detecting the reflected light by the detector; (c) a reference mirror adjustment step of, after the imaging step, controlling the optical distance adjustment unit to move the reference mirror to make the optical distance of the reference light shorter than the optical distance of the reflected light, so that i) the image portion of reflected light from the biological tubular element and the image portion of reflected light from the translucent tube are enlarged and

- ii) an image portion of an artifact caused by the reflected light from the optical part is reduced, whereby fitting the image portion of the artifact within an inside of the image portion of the reflected light from the translucent tube; (d) a magnification adjustment step of, after the reference mirror adjustment step, controlling the image processor to delete the data corresponding to the intensity corresponding to the artifact, and then, decrease size of the image portion of the reflected light from the biological tubular element and the image portion of the reflected light from the translucent tube to size before the enlargement; and (e) a display step of, after the magnification adjustment step, causing the image display to display the image portion of the reflected light from the biological tubular element and the image portion of the reflected light from the translucent tube.
2. The method according to claim 1, further comprising an artifact deletion step of controlling the image processor to delete the reduced image portion of the artifact is executed after the reference mirror adjustment step and before the magnification adjustment step.
3. The method according to claim 1, wherein the optical tomographic imaging apparatus comprises a linear motion unit driven by a motor that moves the optical part of the imaging unit along the translucent tube, the method further comprises: a pull-back imaging step of operating the imaging unit to image the biological tubular element while operating the linear motion unit to move the optical part along the translucent tube is executed after the display step, the magnification adjustment step and the display step are executed after the pull-back imaging step, and the pull-back imaging step, the magnification adjustment step, and the display step are repeated in order.
4. The method according to claim 1, wherein the optical part includes a prism, the prism is a right angle prism that includes one end connected with a first side and a second side connected via a right angle, and one oblique side connecting another end of the first side and another end of the second side, the first side of the right angle prism is optically coupled to a distal end of the optical fiber, the light emitted from the distal end of the optical fiber is reflected by the oblique side and then emitted through the second side, and light reflected by the biological tubular element is incident on the right angle prism from the second side, reflected by the oblique side, and then incident on the optical fiber through the first side.
5. The method according to claim 1, wherein the detector is a dual balanced detector.
6. The method according to claim 1, wherein the converter is an analog-digital conversion unit.
7. The method according to claim 1, wherein the Fourier transform unit is a computer program implemented on the controller.
8. The method according to claim 1, wherein the image processor is a computer program implemented on the controller.
9. The method according to claim 1, wherein the controller is a processor or a computer.
10. The method according to claim 1, wherein the optical distance adjustment unit includes a collimation lens, and in the reference mirror adjustment step (c), the optical distance adjustment unit is controlled to move the reference mirror toward the collimation lens to make the optical distance of the reference light shorter than the optical distance of the reflected light.
-